University of Pittsburgh | School of Public Health

More statistical learning tools
- Tree-based methods
- Bagging, Boosting, and Random forests

BIOST 2155

Class 10
November 13th, 2026

Statistical modeling: The **two cultures** (with comments and a rejoinder by the author)

L Breiman - **Statistical** science, 2001 - projecteuclid.org

… There are **two cultures** in the use of **statistical modeling** to reach conclusions from data. One … **model**. The other uses algorithmic **models** and treats the data mechanism as unknown. The …

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[PDF] projecteuclid.org

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Assigned reading (not compulsory)



Leo Breiman 1928-2005

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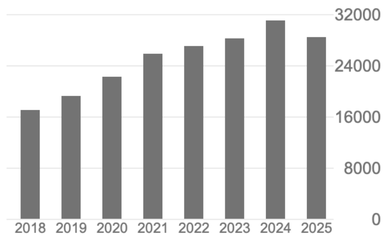
Data Analysis Statistics Machine Learning

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TITLE	CITED BY	YEAR
Random forests L Breiman Machine learning 45 (1), 5-32	167626	2001
Classification and Regression Trees L Breiman, JH Friedman, RA Olshen, CJ Stone CRC Press, New York	~~70583~~ *	1999
Classification and regression trees L Breiman Chapman & Hall/CRC	70583 *	1984
Bagging predictors L Breiman Machine learning 24 (2), 123-140	41191	1996

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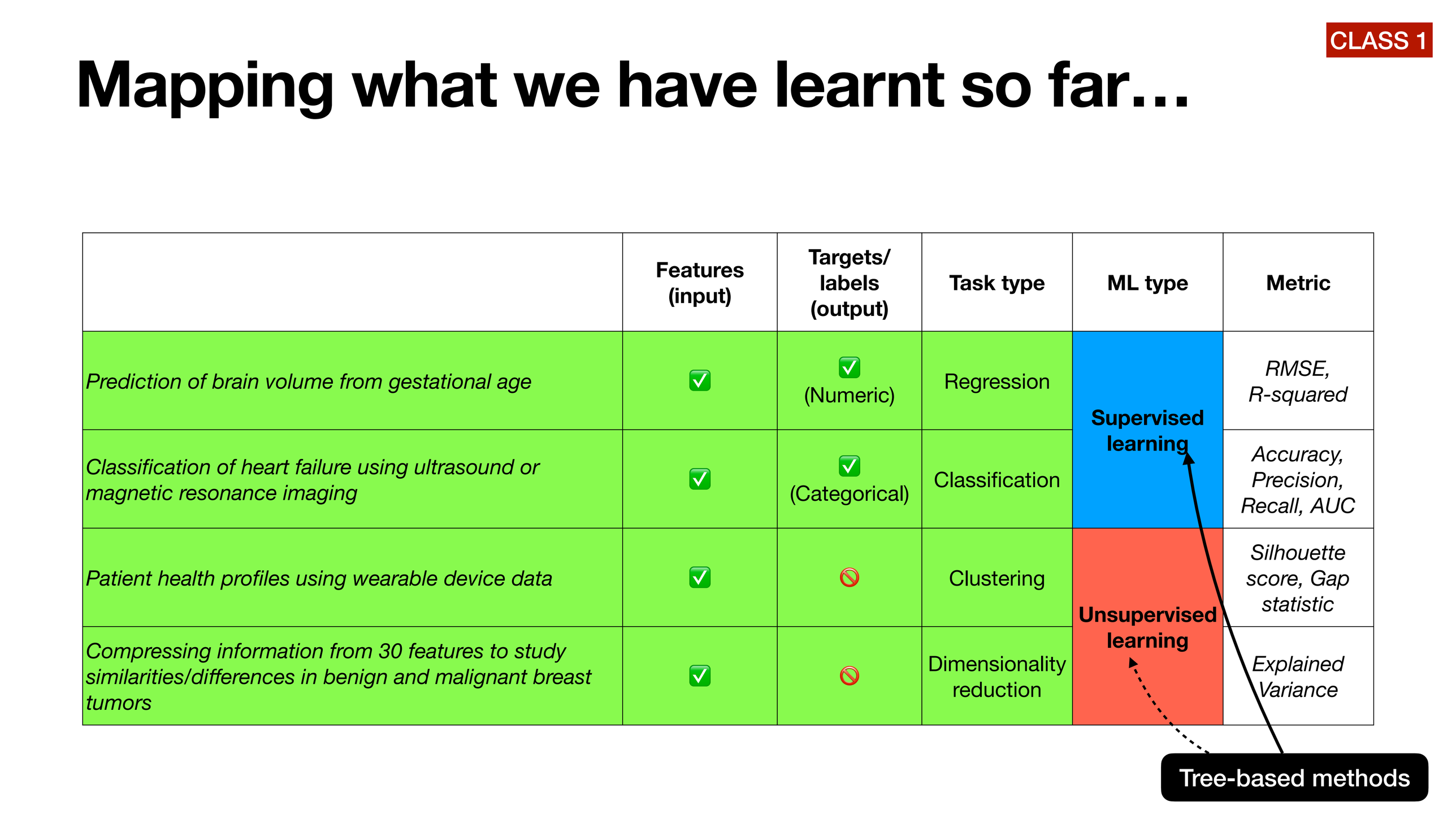


In 2003

CLASS 1

Mapping what we have learnt so far…

	Features (input)	Targets/ labels (output)	Task type	ML type	Metric
Prediction of brain volume from gestational age	✅	✅ (Numeric)	Regression	**Supervised learning**	*RMSE, R-squared*
Classification of heart failure using ultrasound or magnetic resonance imaging	✅	✅ (Categorical)	Classification		*Accuracy, Precision, Recall, AUC*
Patient health profiles using wearable device data	✅	🚫	Clustering	**Unsupervised learning**	*Silhouette score, Gap statistic*
Compressing information from 30 features to study similarities/differences in benign and malignant breast tumors	✅	🚫	Dimensionality reduction		*Explained Variance*



Tree-based methods

Overview

- Classical tree-based methods can be applied for supervised learning: regression and classification tasks.
- Newer tree based methods have been developed for unsupervised learning as well
    - Isolation forests[1]: unsupervised anomaly/outlier detection
    - Unsupervised random forests[2]: applying random forests to derive distances between data points, then use for clustering

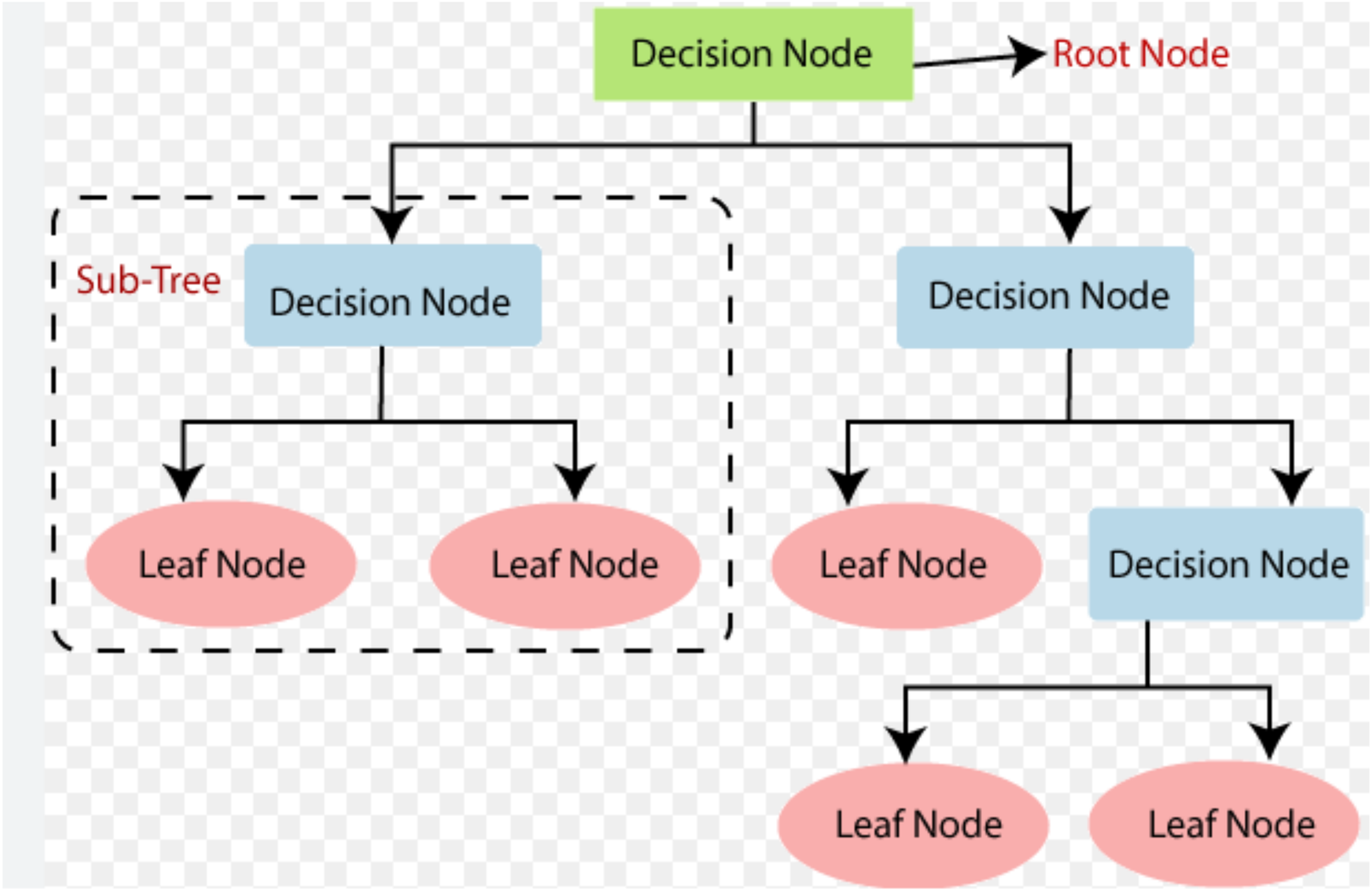
Involve stratifying/segmenting the feature space into some simple regions.

1. F. T. Liu, K. M. Ting and Z. -H. Zhou, "Isolation Forest," 2008 Eighth IEEE International Conference on Data Mining, Pisa, Italy, 2008, pp. 413-422, doi: 10.1109/ICDM.2008.17

2. Shi, Tao, and Steve Horvath. "Unsupervised learning with random forest predictors." Journal of Computational and Graphical Statistics 15.1 (2006): 118-138.

Tree-based methods

Terminology

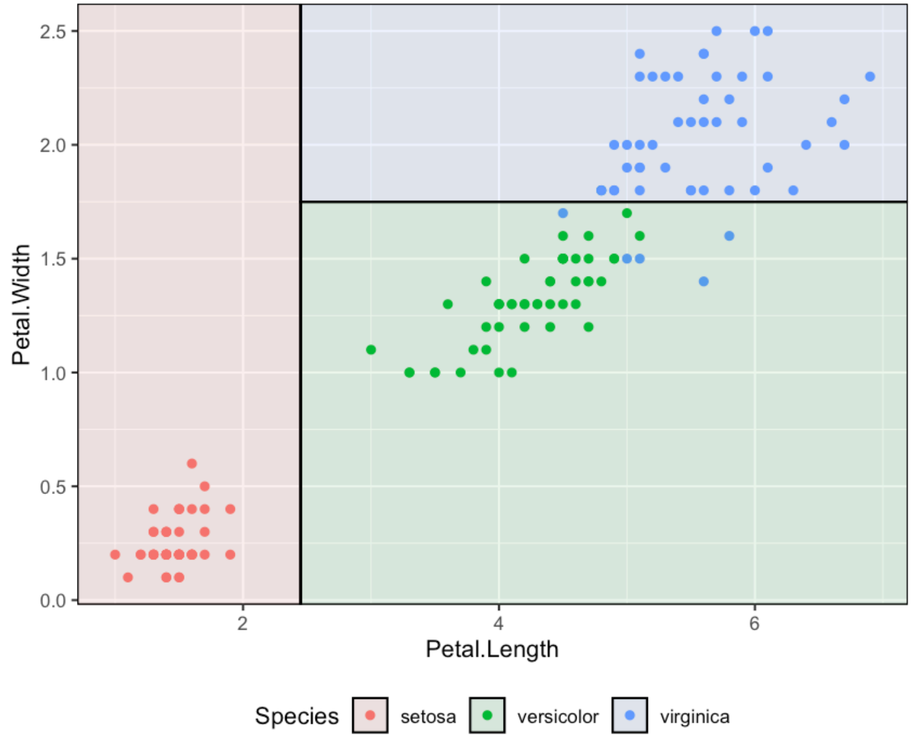


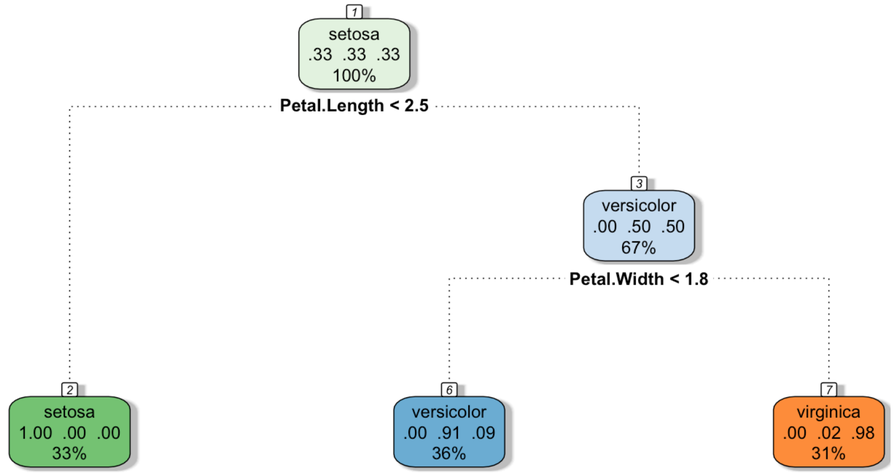
Tree-based methods

Strengths, weaknesses, and scope for improvement…

- **Tree-based methods are simple and useful for interpretation.**
- However they typically are not competitive with the best supervised learning approaches in terms of prediction accuracy.

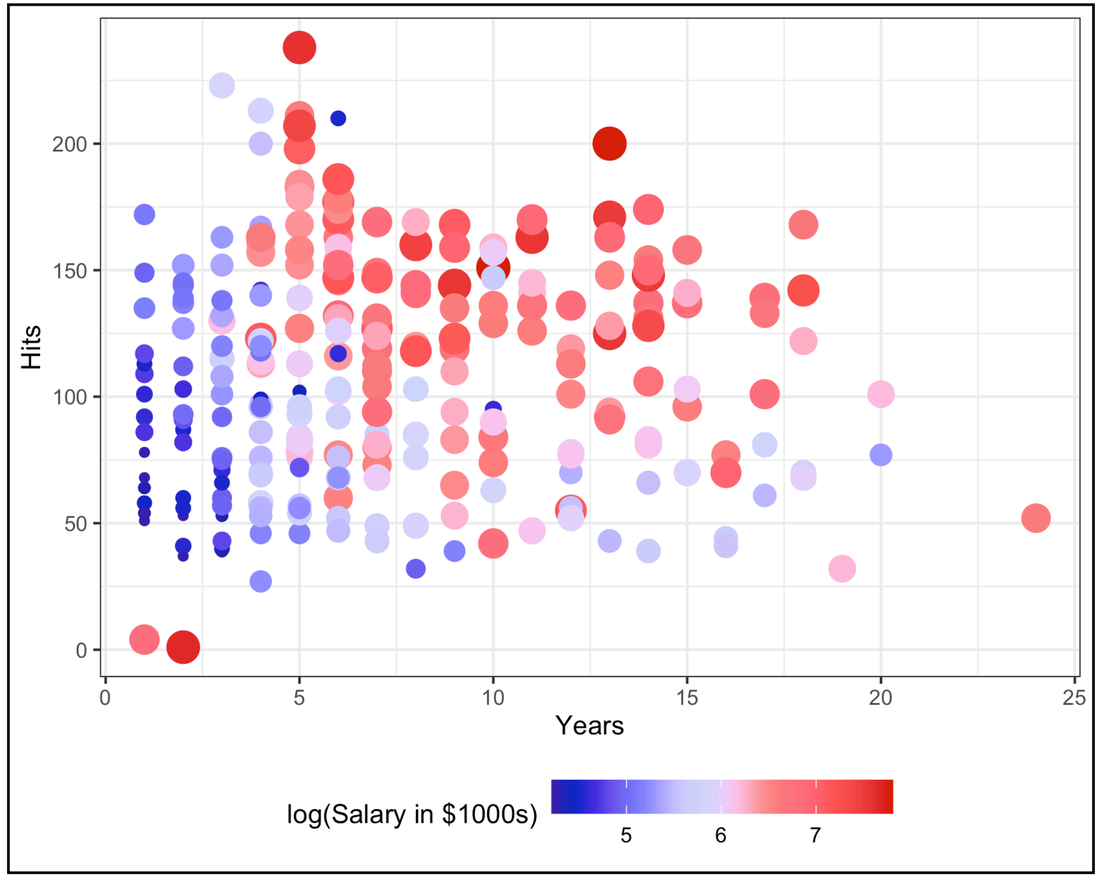
- **Hence we also discuss bagging, random forests, and boosting. These methods grow multiple trees which are then combined to yield a single consensus prediction.**
- Combining a large number of trees can often result in dramatic improvements in prediction accuracy, at the expense of loss of interpretation.

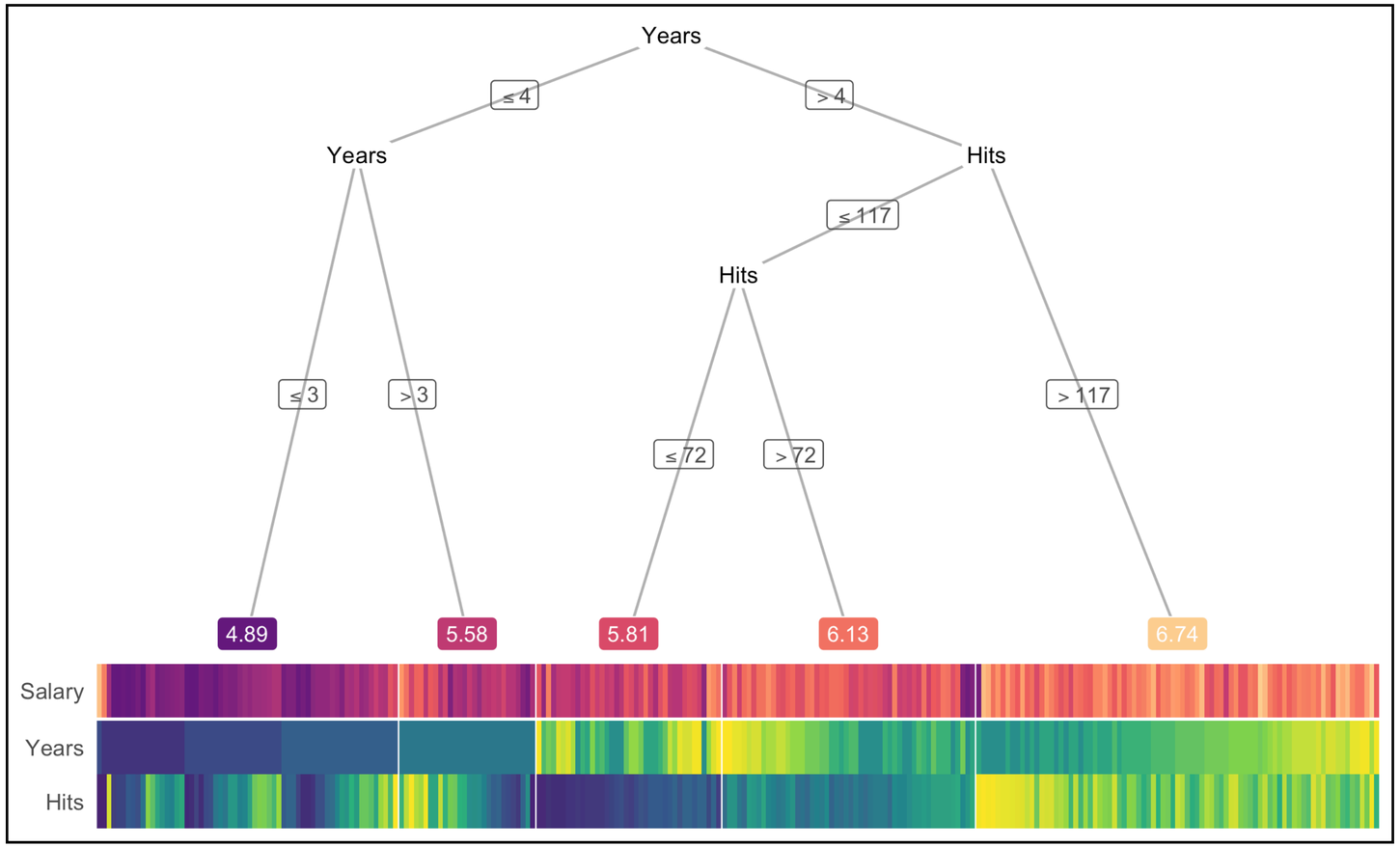




Regression trees

Predicting Baseball Players' Salaries Using Regression Trees





Prediction via Stratification

Details of the regression tree-building process

- We divide the predictor space — that is, the set of possible values for $X_1, X_2, \ldots, X_p$ — into J distinct and non-overlapping regions $R_1, R_2, \ldots, R_J$
- For observations in the region R_j, we make the same prediction: the mean of the response for all training data in R_j:

$$\hat{y}_{R_j} = \text{ave}_{i \in R_j} y_i$$

- **How to define regions $R_1, R_2, \ldots, R_J$?**
- **How to quantify model performance?**

Prediction via Stratification

More details of the regression tree-building process

- The regions $R_1, R_2, \ldots, R_J$ could have **any** shape.
- Divide the predictor space into p**-dimensional hypercubes** for simplicity and for ease of interpretation.
    - For $p = 1$: intervals; for $p = 2$: rectangles; for $p = 3$ cubes….
- In specialized circumstances, can think of concentric circles, spheres, hyperspheres: epidemiology, climate sciences, etc.

Optimization problem: find regions $R_1, R_2, \ldots, R_J$ that minimize:

$$RSS = \sum_{j=1}^{J} \sum_{i \in R_j} \left(y_i - \hat{y}_{R_j} \right)^2$$

Prediction via Stratification

EVEN MORE details of the regression tree-building process

- Computationally infeasible to consider every possible partition of the feature space into J boxes.
- Take a **top-down**, **greedy** approach: **recursive binary splitting**.
    - **Top-down**: begins at the top of the tree and then successively splits the predictor space; each split is indicated via two new branches further down on the tree.
    - **Greedy:** at each step, the best split is made at that particular step, rather than looking ahead and picking a split that will lead to the best tree in some future step.

Optimization problem: find regions $R_1, R_2, \ldots, R_J$ that minimize:

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Prediction via Stratification

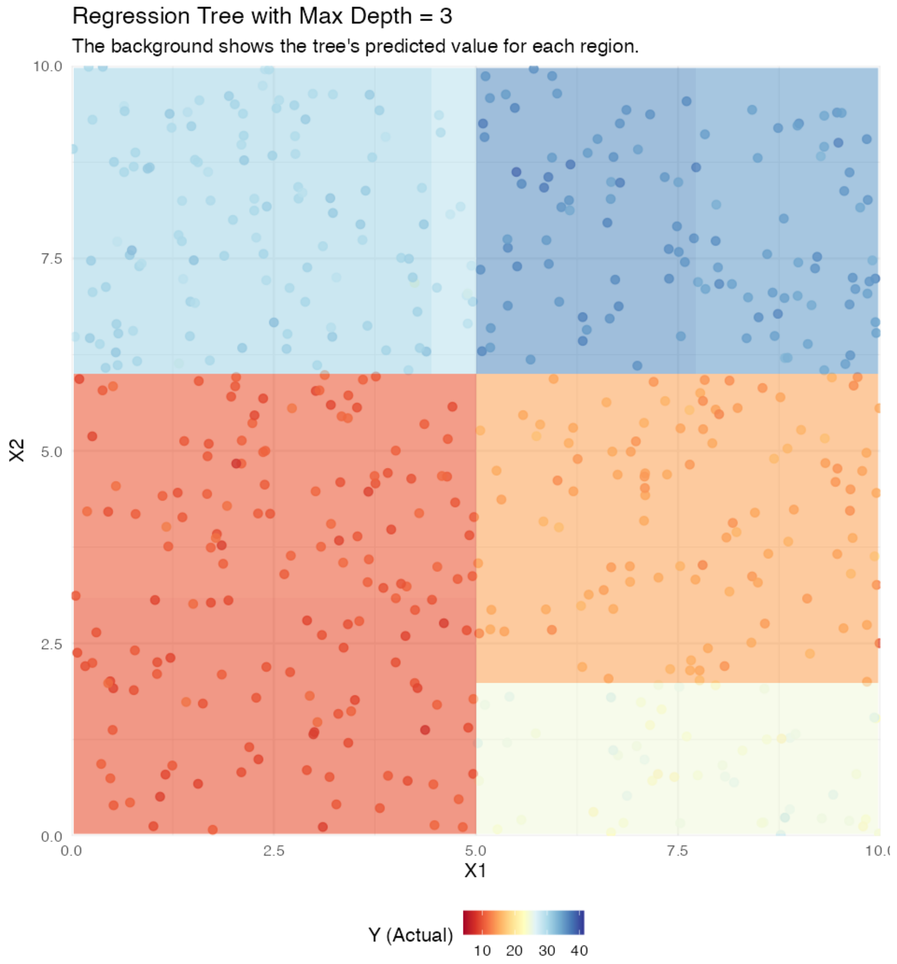
EVEN MORE (x2) details of the regression tree-building process

- Step 1: select X_j and cut-point s such that splitting the predictor space into the regions $R_1(j,s) = \left\{X \,|\, X_j < s\right\}$ and $R_2(j,s) = \left\{X \,|\, X_j \geq s\right\}$ leads to the greatest possible reduction in RSS.
- Step 2: look for the best predictor and best cut-point in order to split the data further so as to minimize the RSS within each of $R_1(j,s)$ and $R_2(j,s)$.
    - Instead of splitting the entire space, split **only one** of $R_1(j,s)$ and $R_2(j,s)$: now have three regions.
- Step 3: we look to split **only one** of these three regions further.
- Repeat until a **stopping criterion** is reached: until no region contains more than five observations (‘minimum points per node’) or until a tree of some pre-specified ‘depth’ has been constructed.

Prediction via Stratification

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Pruning a tree

Trees struggle a lot with overfitting

- Produces good predictions on the training set, but is likely to overfit the data, leading to poor test set performance - because the resulting tree T_0 might be too complex.
- A smaller tree T_1 with fewer splits might lead to lower variance and better interpretation at the cost of a little bias.
- Mitigation strategy: grow a very large tree T_0, and then **prune** it back in order to obtain a subtree. Cost complexity pruning — also known as weakest link pruning — is used to do this.



Large tree with low bias but (potentially) high variance because of overfitting



Pruned tree - more bias but lower variance, better overall MSE

Pruning a tree

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1. Start with T_0 and specify tuning parameter α
2. Find $T_1 \subset T_0$ that minimizes

$$\sum_{m=1}^{|T_1|} \sum_{i:x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T_1|$$

$|T_1|$: number of split-up regions in T_1

R_m: split-up regions

$\hat{y}_{R_m}$: mean of training observations in R_m

Choose α using k-fold CV.

Larger α will give you smaller trees!

Prediction via Stratification

Details of the *classification* tree-building process

- Very similar to a regression tree, except that it is used to predict a qualitative response rather than a quantitative one.
- Predict that each observation belongs to the **most commonly occurring class** of training observations in the region to which it belongs.
- Need alternatives to RSS within R_m: we define $\hat{p}_{mk}$ = proportion of training observations in R_m that are from the k-th class

Misclassification rate: $E = 1 - \max_k \hat{p}_{mk}$

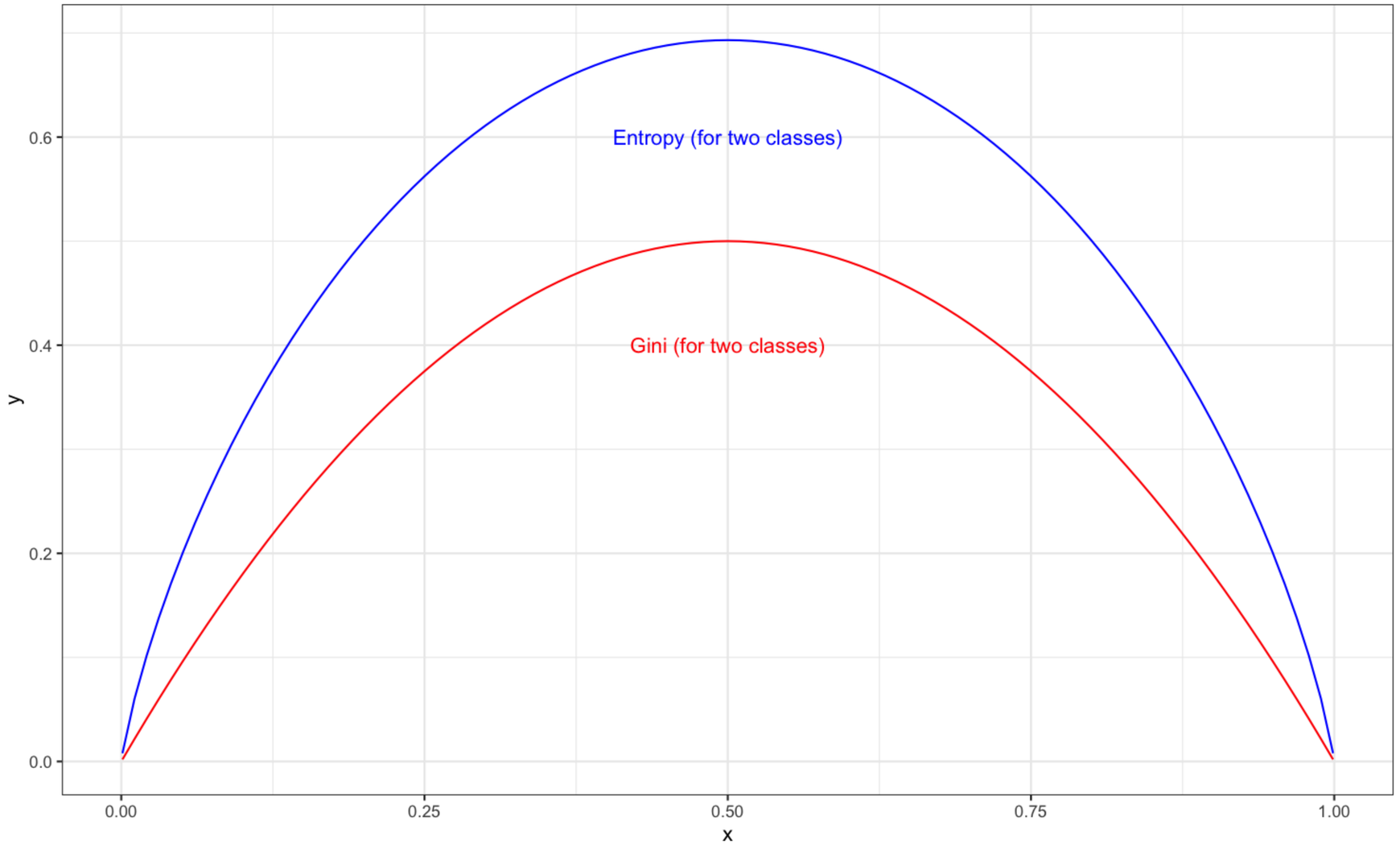
Gini index: $G = \sum_{k=1}^{K} \hat{p}_{mk}\left(1 - \hat{p}_{mk}\right)$

Entropy: $D = -\sum_{k=1}^{K} \hat{p}_{mk} \log(\hat{p}_{mk})$

Not sufficiently sensitive

Measures of impurity in node

We will still make use of recursive binary pruning, using G or D



Misclassification rate: $E = 1 - \max_k \hat{p}_{mk}$

Gini index: $G = \sum_{k=1}^{K} \hat{p}_{mk}\left(1 - \hat{p}_{mk}\right)$

Entropy: $D = -\sum_{k=1}^{K} \hat{p}_{mk} \log(\hat{p}_{mk})$

Not sufficiently sensitive

Measures of impurity in node

Advantages and Disadvantages of Trees

- Trees are very easy to explain to people: easier to explain than linear regression!
- Decision trees more closely mirror human decision-making than do the regression and classification approaches.
- Trees can be displayed graphically, and are easily interpreted even by a non-expert (especially if they are small).
- Trees can easily handle qualitative predictors without the need to create dummy variables.

- Trees generally do not have the same level of predictive accuracy as most other regression and classification approaches.
- A small change in the data can cause a large change in the final estimated tree

Opportunity: By aggregating many decision trees, the predictive performance of trees can be substantially improved….

Improvement 1: BAGging

BAG = Bootstrap Aggregation

- General-purpose procedure for reducing the variance of ANY statistical learning method; particularly useful in the context of decision trees.
- Given a set of n independent observations $Z_1, Z_2, \ldots, Z_n$ each with variance σ^2, the variance of the mean $\bar{Z}$ is σ^2/n - which means averaging a set of observations reduces variance…but we don't have multiple training sets!

- Bootstrap (sample rows with replacement) B different training sets.
- Train models on each bootstrapped set, call them $\hat{f}_b(\boldsymbol{x}),\ 1 \leq b \leq B$

Aggregate all models $\hat{f}_{\mathbf{bag}}(\boldsymbol{x}) = B^{-1} \sum_{b=1}^{B} \hat{f}_b(\boldsymbol{x})$

Classification trees: get $\boldsymbol{x}$, record $\hat{f}_b(\boldsymbol{x})$, take a majority vote.

Improvement 1: BAGging

Out-of-Bag error estimation

- Recall that the **key to bagging** is that trees are **repeatedly fit to bootstrapped subsets of the observations**.
  - One can show that on average, each bagged tree makes use of around two-thirds of the observations….recall 0.632 bootstrap!
- **The remaining one-third of the observations not used to fit a given bagged tree are referred to as the out-of-bag (OOB) observations. No extra computation cost!**
- We can predict the response for the ith observation using each of the trees in which that observation was OOB.
  - **This will yield around $B/3$ predictions for the ith observation, which we average.**
- This estimate is essentially the LOO cross-validation error for bagging, if B is large.

Improvement 2: Boosting

Boost model performance by learning from prior errors

- Boosting is similar to bagging - we bootstrap B training samples of size n
- Bagging involves creating multiple copies of the original training data set using the bootstrap, fitting a separate decision tree to each copy, and then combining all of the trees in order to create a single predictive model.
- **In boosting, trees are grown sequentially: each tree is grown using information from previously grown trees.**

Boosting for regression:

1. Set $\hat{f}(\boldsymbol{x}_i) = 0$ and $r_i = y_i$, $1 \leq i \leq n$
2. For $1 \leq b \leq n_B$:
    1. Fit tree $\hat{f}_b$ with $d + 1$ nodes with $\boldsymbol{x}$ and r
    2. Update $\hat{f} \leftarrow \hat{f} + \lambda \hat{f}_b$ where λ shrinks $\hat{f}_b$
    3. Update $r_i \leftarrow r_i - \lambda \hat{f}_b(\boldsymbol{x}_i)$
3. Output $\hat{f} = \lambda \sum_{b=1}^{n_B} \hat{f}_b$

n_B: number of boosted trees

λ: shrinkage parameter.

How much does $\hat{f}_b$ contribute to $\hat{f}$

d: number of nodes in the tree.

How granular is $\hat{f}_b$?

Improvement 3: Random forests

Bagging + decorrelating trees

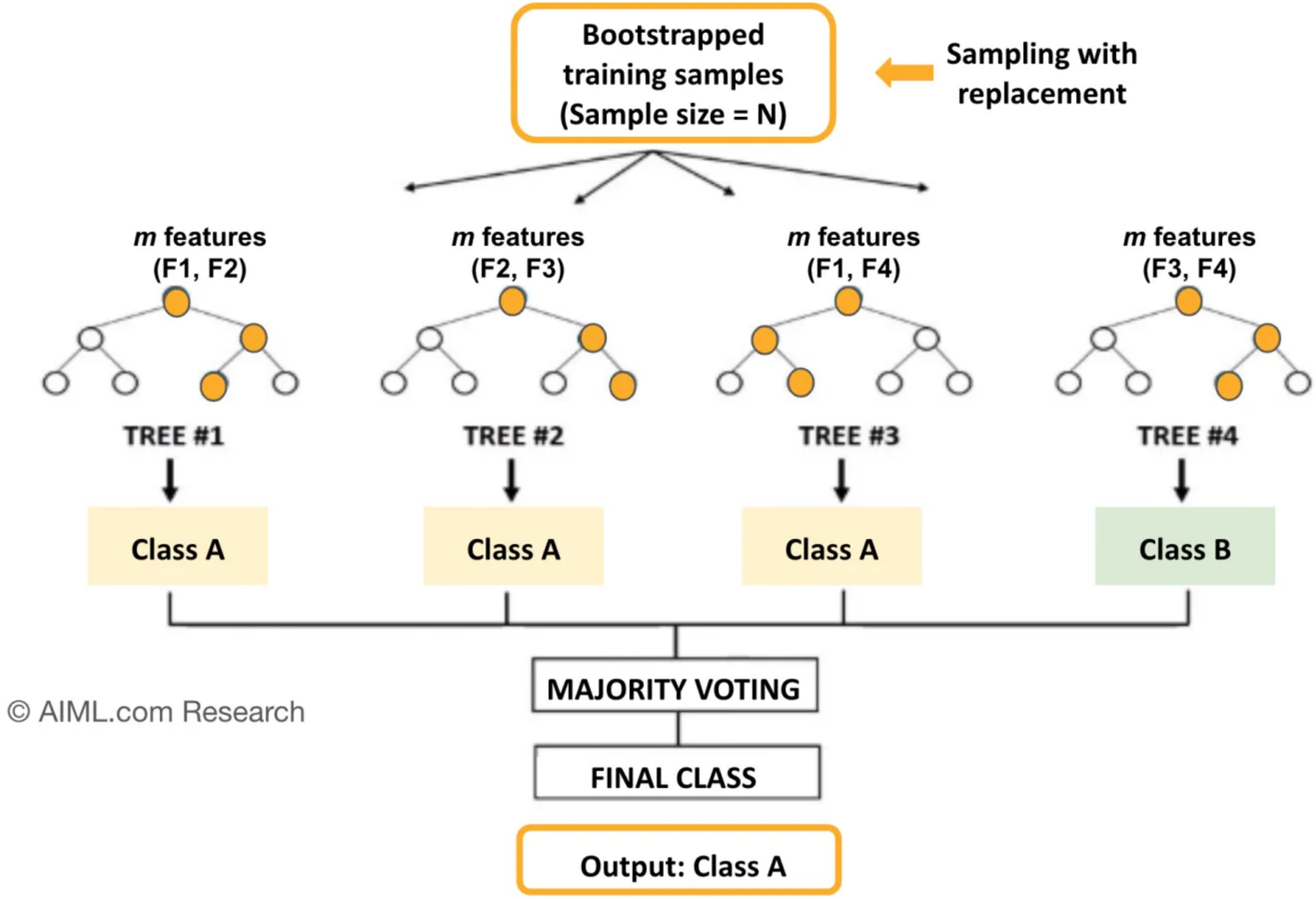
- Challenge with bagging: when predictions are correlated, then the gain is limited (if one model $\hat{f}_b$ makes a mistake, other model $\hat{f}_{b'}$ likely to make mistake too…)
- So we should probably allow for variation in models as well - we do this by allowing different subsets of covariates/features within each base model.
- When building trees, each time a split in a tree is considered, a random selection of m predictors is chosen as split candidates from the full set of p predictors. The split is allowed to use only one of those m predictors.
- Typically $m \approx \sqrt{p}$

Improvement 3: Random forests

Bagging + decorrelating trees

Training Data
(Sample size, N= 6, No. of features, F = 4)

F1	F2	F3	F4	Y
2.1	0	400	-9	A
3.0	1	890	-42	B
2.2	1	929	0	B
4.0	0	324	-23	A
3.5	1	333	-15	A
6.0	0	215	-9	A

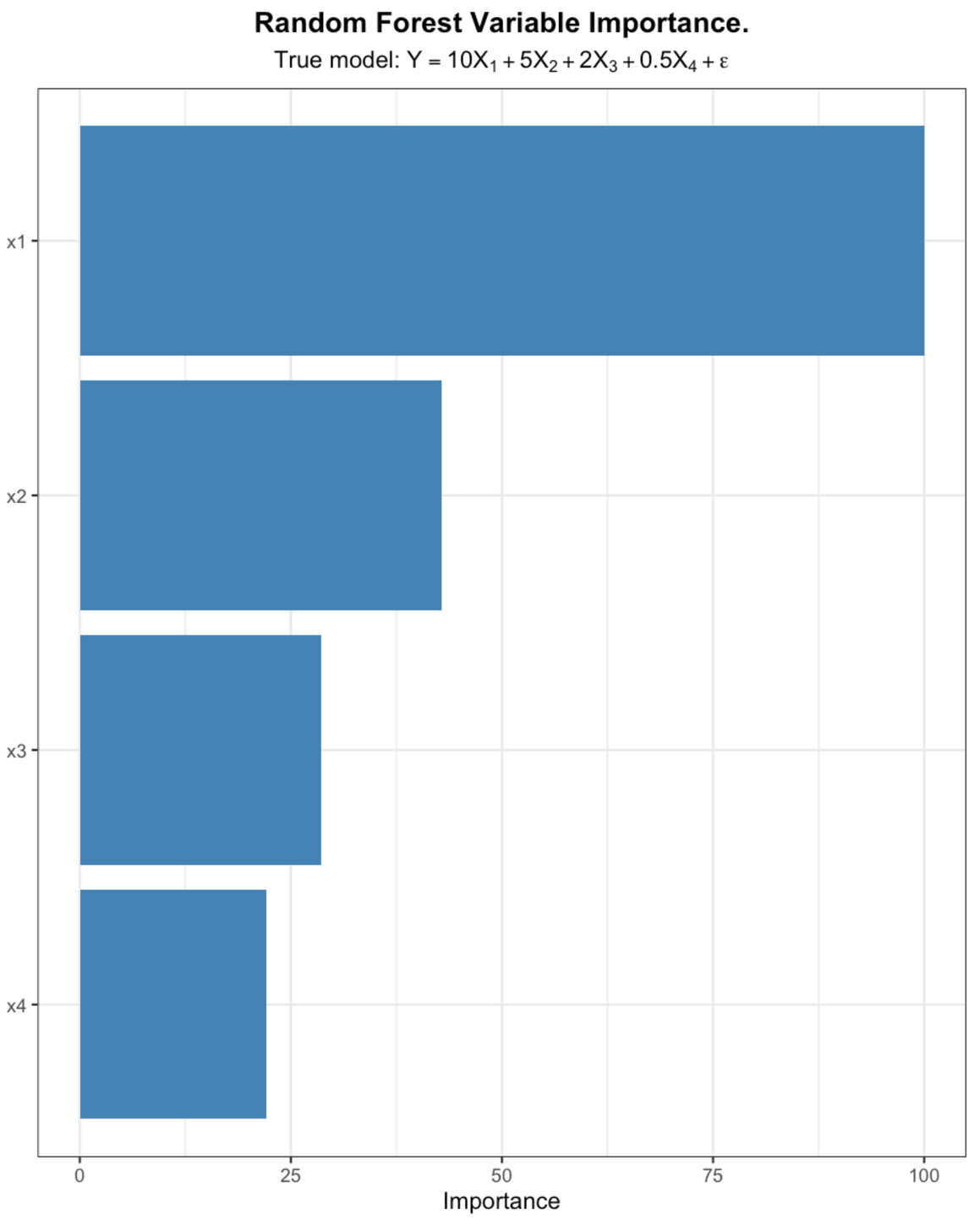


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Variable importance

- For bagged/RF regression trees, we record the total amount that the RSS is decreased due to splits over a given predictor, averaged over all B trees. A large value indicates an important predictor.
- For bagged/RF classification trees, we add up the total amount that the Gini index/ entropy is decreased by splits over a given predictor, averaged over all B trees

Can do variable importance for boosted trees as well using permutation importance.



Post-class debrief

Today we discussed…

- Decision trees are simple and interpretable models for regression and classification
- However they are often not competitive with other methods in terms of prediction accuracy

- Bagging, random forests and boosting are good methods for improving the prediction accuracy of trees.
- They work by growing many trees on the training data and then combining the predictions of the resulting ensemble of trees.

- The latter two methods— random forests and boosting— are among the state-of-the-art methods for supervised learning.
- However their results can be difficult to interpret.


That's all Folks!